

YUEZHEN (LILY) DONG

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RELEVANT SKILLS

Languages	Python, Java, C/C++, TypeScript/JavaScript, SQL, HTML/CSS, Ruby, Bash
Frameworks	React/Next.js, Ruby on Rails, Spring Boot, Node/Express, Three.js, Flask, Arduino, GraphQL
Developer Tools	AWS, Docker, Kubernetes, Git, Maven, Linux, PostgreSQL, AutoCAD, VSCode, IntelliJ

EDUCATION

University of Waterloo - Faculty of Engineering	Expected Graduation MAY 2028
BASc in Electrical Engineering	Waterloo, ON, CAN

WORK EXPERIENCE

Software Engineering Intern	SEP 2025 – DEC 2025
React, TypeScript, Ruby on Rails, GraphQL	Shopify

- POS Channel: shipped “Sell in person” capability across Admin & POS (**React+TypeScript, Rails/GraphQL, Vault-backed flows**); designed end-to-end onboarding/activation used by **millions+** of merchants, cutting time-to-first-sale by **35%** and lifting POS activation conversion by **20+ pp**.
- Terms of Service (TOS): implemented global TOS acceptance/enforcement across Admin/POS (backend services, gating, audit trails); staged rollout to **100%** of stores via feature flags, achieving **99%+** compliant usage and reducing support tickets by **60%**.
- Drove **50+** of **full-stack** PRs with production quality: typed APIs, integration/unit tests (**95%** coverage), analytics/instrumentation and dashboards, runbooks, and safe deploys; partnered with PM/design to deliver on schedule.

Web Developer Coop	JAN 2025 – APR 2025
Java, Spring Boot, Maven, Apache POI, Git	AGF Investments

- (**In production — company-wide**) Independently developed a **Java-based** ScoreCard application with file archiving, large XLSX handling via **Apache POI streaming**, data validation/transformation, and a properties-based configuration system; now used in **daily reporting workflows across 4 departments**, saving **20+ staff hours/week**.
- (**In production — company-wide**) Delivered an end-to-end **Spring Boot** reporting pipeline to ingest XLSX, filter by date/time, archive inputs, generate Excel/CSV outputs, and automated email distribution under **Maven/Git**.
- Built **Pega** scenario tests and **Selenium** UI/service tests covering 90% of core business flows, reducing regression defects by **40%** and enabling releases to ship **2x faster**.

Website Developer	DEC 2020 – DEC 2023
Java, HTML/CSS, JavaScript	Beyond the Wards / Kindness Catalogue

- Designed and deployed a dynamic website using **Java** backend and **HTML/CSS, JavaScript** frontend, ensuring mobile responsiveness and smooth interaction; maintained uptime of **99.9%** and supported **5,000+ monthly visitors**.

PERSONAL PROJECTS

Ecoland: AI Ecosystem Simulator
Java, JavaFX, OOP, AI, Genetic Algorithms

- Built a **Java** simulator with custom **AI algorithms**, pathfinding, and genetic algorithms to model species interactions.
- Employed **concurrency**, **data structures**, and **OOP** principles for large-scale simulations, enabling real-time monitoring, performance tuning, and modular feature expansions.

3D Data Sandbox
Electron, Node.js, Three.js, PapaParse, JavaScript

- Built a cross-platform **Electron** desktop application using **Three.js** and **Node.js/PapaParse** for offline 3D visualization, parsing, and analysis of local CSV/JSON datasets.
- Implemented scatter, grid, and K-means layouts with outlier detection, correlation lines, and user-friendly UI controls for filtering, highlighting, and exporting.

Puppy Portal Puzzle: Time-Trotting Dogventure
Python, Pygame, Pathfinding, Time-Manipulation, Puzzle Architecture

- Developed a controlled **time-travel** mechanic with simultaneous clones, integrating **BFS pathfinding** for AI enemies.
- Structured the game using **Python OOP** and event-driven architecture, implementing multi-layer switch logic, portal systems, and crate interactions while ensuring seamless performance under increasing complexity.